Response to Reviewer Comments

QMOF-Rec: A Multi-Objective Recommender-System Framework for Quantum-Chemical MOF Candidate Screening

We thank the reviewer for the careful and constructive assessment. We completed a final consistency pass across the source code, manuscript, supplementary material, figures, and response package. The revision now separates retrieval missingness from numerical reranking missingness, moves availability masks into the general mathematical formulation, excludes unavailable void fraction from active numerical objectives, reports query-stratified repeated-seed results, adds candidate-pool-size sensitivity for the available artifacts, and clarifies the LEA no-diversity result without claiming general equivalence.

1. Missing descriptors and void fraction

Reviewer comment: Missing descriptors and void fraction.

Response: We agree that missing descriptors must not be interpreted as physical zero values. The revised code uses explicit availability masks and masked weighted scoring/distances. Void fraction is unavailable for 0/20,372 records and is excluded from active numerical ranking, diversity, remapping, radar plots, and frontend metric plots.

Changes in manuscript/code: Manuscript: Abstract, Problem Formulation, Experimental Setup, Results. SI: Descriptor Availability and Masking. Code: backend/app/recommendation/objective_utils.py, feature_extractor.py, hybrid_ranker.py, lea_optimizer.py, material_similarity.py, novelty_ranker.py, recommendation_pipeline.py.

2. Retrieval versus reranking

Reviewer comment: Retrieval versus reranking.

Response: We separated text-embedding retrieval from numerical reranking. The deployed retrieval layer uses SentenceTransformer embeddings of metadata text templates in a FAISS L2 index; it does not search over the numerical objective vector. Unavailable fields are serialized as unavailable metadata, while reranking uses objective vectors plus masks.

Changes in manuscript/code: Manuscript: Retrieval-Augmented Systems and Vector Search. SI: Retrieval Versus Reranking Missingness. Code: backend/scripts/build_vector_index.py, backend/app/rag/retriever.py, backend/app/rag/vector_store.py.

3. LEA no-diversity and WeightedSum

Reviewer comment: LEA no-diversity and WeightedSum.

Response: We rechecked the saved repeated-seed metric artifact. LEA no diversity and WeightedSum have identical saved Rel@5 and diversity values for all 50 query/seed rows. The saved metrics CSV does not contain top-K identifiers, so top-K identity and ordering cannot be verified from that artifact. We therefore retain the conservative wording: configuration-specific empirical identity, not a general equivalence claim.

Changes in manuscript/code: Manuscript: Ablation and Graph-Aware Recommendation Results. SI: metric-level LEA no-diversity check. Artifact: reviewer_revision_artifacts/lea_no_diversity_metric_identity.csv.

4. Five queries x ten seeds

Reviewer comment: Five queries x ten seeds.

Response: We agree that the 50 query/seed rows are nested within five query scenarios and are not 50 independent scientific tasks. The supplement now reports query-stratified results and a direct variance decomposition showing that between-query variance dominates within-query seed variance.

Changes in manuscript/code: Manuscript: Statistical Analysis and Additional Robustness Scope. SI: Query-stratified repeated-seed table and variance-decomposition table. Artifacts: query_stratified_seed_results.csv and variance_decomposition.csv.

5. Candidate-pool-size trade-off

Reviewer comment: Candidate-pool-size trade-off.

Response: The available saved artifacts contain LEA pool-size variants for 50, 100, and 200 candidates. We report these values and state that pool size 100 is retained as the predefined main experimental setting to maintain consistency with the primary evaluation, not because the sensitivity analysis establishes it as optimal. Pool sizes 50--200 produce broadly similar quality, while runtime increases moderately with pool size.

Changes in manuscript/code: Manuscript: Experimental Setup and Ablation discussion. SI: Candidate-pool-size sensitivity table. Artifact: candidate_pool_size_summary.csv.

6. Table 4 versus Table 7

Reviewer comment: Table 4 versus Table 7.

Response: We traced the difference to protocol: Table 4 is the saved primary five-scenario single-run ranking artifact, while Table 7 is the historical repeated-seed ablation protocol generated with the earlier configurable LEA script. The manuscript now states this distinction explicitly and keeps the abstract values tied to the primary single-run artifact.

Changes in manuscript/code: Manuscript: Recommendation Performance and Ablation Results. SI: Repeated-seed scope.

7. Mathematical equations

Reviewer comment: Mathematical equations.

Response: We moved the availability-mask definitions into the general formulation and updated relevance, balance, diversity, and LEA remapping equations to use masks. The balance term is now implemented as evenness over observed active objectives rather than as mean objective quality. We also distinguish masked cosine similarity, used for auxiliary material-similarity scoring, from the masked distance used for LEA diversity and candidate remapping.

Changes in manuscript/code: Manuscript: Problem Formulation; LEA-Guided Multi-Objective Reranking; Algorithms.

8. Figure S3 and table presentation

Reviewer comment: Figure S3 and table presentation.

Response: We renamed and regenerated Figure S3 as an observed physical descriptor comparison using band gap and density only. The caption now states that semantic and stability scores are excluded from this physical-descriptor plot and that void fraction/porosity is excluded because void fraction is unavailable. Main tables also indicate best and second-best values and specify whether higher or lower is better.

Changes in manuscript/code: Manuscript: Table 4 and Table 7 captions. SI: Observed Physical Descriptor Means: All Records Versus Selected Top-K Portfolios. Figure: figures/figure_s3_all_vs_selected_active_descriptors.png.

9. References, terminology, and AI-use statement

Reviewer comment: References, terminology, and AI-use statement.

Response: We added Seko et al. and Qu et al., corrected the KadiAssistant author/title metadata, separated Bayesian optimization from evolutionary/swarm methods, standardized recommender terminology, expanded major abbreviations, reduced instruction-like phrasing, and updated the generative-AI disclosure to include language editing, consistency checking, code-review support, figure-preparation support, and response-to-review preparation.

Changes in manuscript/code: Manuscript: Related Work, Future Work, Declarations, Bibliography. Bibliography: references.bib.

10. Repository and reproducibility

Reviewer comment: Repository and reproducibility.

Response: The local repository branch contains the descriptor-mask implementation, corrected masked balance score, focused regression tests, and reproducibility docs. A protocol_audit.csv and final_configuration.json were also added to the manuscript folder to identify which saved artifacts produced each table/figure. Pushing or confirming a clean GitHub clone remains dependent on GitHub credentials, which were unavailable in the local environment.

Changes in manuscript/code: Repo: README.md, REVISION_CHANGELOG.md, REPRODUCTION_COMMANDS.md, REVIEWER_COMMENT_MATRIX.csv, FINAL_REVISION_AUDIT.md. Manuscript artifacts: protocol_audit.csv and final_configuration.json. Commits created during revision: beb19d7e, 638ccd5a, and f8b52c00.

11. Missing-data ablation

Reviewer comment: Missing-data ablation.

Response: A full zero-fill versus neutral-fill versus masked-scoring ablation was not added because the saved repeated-seed metrics do not include top-K identifiers needed to count changed top-K lists, and a full rerun under the revised masking protocol would create a new primary-results protocol. Instead, the preferred masked method was implemented and tested.

Changes in manuscript/code: SI: Missing-Data Ablation Status. Code tests: backend/tests/test_missing_data_masks.py.

Final technical audit: We preserved historical numerical results transparently rather than fabricating a mask-aware rerun. The new protocol_audit.csv maps the Abstract, Table 4, Table 7, statistical summaries, candidate-pool sensitivity, LEA-no-diversity check, Figure S3, and current code tests to their exact protocols and artifacts.

Final pre-submission audit: We corrected the Seko citation context, defined graph convolutional network (GCN) at first use, changed the retrieval formulation to FAISS L2 distance to match the live vector store, added an explicit archived-results versus post-review mask-aware-code protocol statement before the results section, changed the Table 7 LEA baseline note to the historical configurable LEA parameters, improved the abstract missing-descriptor test wording, and moved Algorithms 1--3 plus the detailed RAG/LLM rubric and hallucination-flag definitions to the supplementary material. The main manuscript now builds as 37 pages rather than 39 pages.

Final polish: We clarified that Figures 4--6 summarize the archived evaluation while Figure 7 illustrates the revised active objective representation, defined GAT in the abstract, defined MAE/RMSE/R^2/Spearman before Table 6, made the Table 4 TOPSIS hypervolume entry math-bold, changed Contribution 5 to query-blocked statistical robustness analyses, and revised the Table 5 caption to refer to the saved offline evaluation artifact.

Final presentation cleanup: We split the crowded Table 7 NDCG/Diversity headers over two lines, removed the internal Table B label from the RAG/LLM results caption, and changed the Table 3 caption to refer to the saved offline QMOF-Rec evaluation artifact.

Verification

Focused backend tests: PYTHONPATH=backend pytest -q backend/tests/test_missing_data_masks.py passed (8 tests).

Main manuscript build: latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex passed.

Supplementary build: latexmk -pdf -interaction=nonstopmode -halt-on-error supplementary_material.tex passed.

Final manuscript PDFs produced: main.pdf and supplementary_material.pdf. Main and supplementary builds passed after the final pre-submission consistency edits, final polish pass, and presentation cleanup. Duplicate manuscript PDF variants and LaTeX build byproducts were removed after successful compilation.